A BM QUICK START

Your first game in bm

Make a 2D game with a title, three stages, an ending and saved records on the RGB30 and the Pi Zero W.

Contents

Before you start 3

Make the game on the RGB30 4

Make it yours 5

Play it from the menu 6

On the Pi: change the code with bm Code 7

Keep it at 60 fps 8

Where to go next 9

Open points 10

README

What this is. A one-page guide that takes a developer who has never used bm from the menu to a complete 2D game with the built-in suite.

How to use it. Follow the seven numbered chapters in order. Every screen action is shown as a key or menu step, followed by what appears on screen.

What is missing. Gaps waiting for upcoming templates, tables, or hardware checks are tagged with highlighted TO FILL markers.

Before you start

bm creates cartridges directly on the console or from source.

You need	For
PowKiddy RGB30 with a bm card	making the game and playing it on the handheld
USB keyboard	comfortable typing (the gamepad uses the on-screen keyboard)
Raspberry Pi Zero W with its bm card	testing and modifying code on the Pi (optional, chapter 5)
A PC with the bm repository	copying files between cards or over the network

You will build a 2D platformer with a title screen, 3 stages, an ending, and a saved high score.

Make the game on the RGB30

1. From the main menu, go to Dev and open bm SDK.
2. Press `Ctrl+N` to open `new project from a template`, pick **Stages 2D**, and press Enter.
3. Press `F5` to test the game; press Start + Select to return to the editor.
4. Press `Esc` to open the project menu, choose `Save as...`, enter `MYGAME.BME`, and press Enter.

[TO FILL: the Stages 2D template is not in bm SDK (RGB30) yet: check these steps when it is]

Make it yours

1. Press `F2` to open the code editor, and change the title text in the Lua source.
2. Press `F3` twice to open the map editor: stage n occupies 45 rows starting at row (n-1)×48 across the first 128 columns; tile 7 is the exit.
3. Press `Ctrl+S` to save your changes.

[TO FILL: the Stages 2D template is not in bm SDK (RGB30) yet: check these steps when it is]

Play it from the menu

1. In bm SDK, press `Ctrl+Shift+B` : the compiled `.b16` goes to `/bm/` and appears in the Games tab.
2. Press `Ctrl+B` to build `MYGAME.BM` in `/carts/` , with the message `(the Games tab lists bm/)` : this is the file for the Pi.

On the Pi: change the code with bm Code

1. Copy `MYGAME.BME` and `MYGAME.BM` from the `carts/` folder of the RGB30 card into the `carts/` folder of the Pi card.
2. In the Pi main menu, go to Dev and open Code.
3. Press `Ctrl+O`, choose `/carts/MYGAME.BME`, and press Enter.
4. Modify the code and press `F5` to test: if an error occurs it jumps to the line, and `F9` explains it.
5. Press `Ctrl+S` to save. The Pi SDK does not have the Stages 2D template.

[TO FILL: a project made on the RGB30 opened in bm Code on the Pi: not tried yet] **[TO FILL: Ctrl+Q, the help on the function under the cursor in bm Code: not there yet]**

Keep it at 60 fps

1. Keep each frame within the 16.7 ms budget to maintain a steady 60 fps.
2. Press `F11` to toggle the developer overlay and check frame times.
3. Use `profile(true)` and `profile()` to measure routines; see `docs/PERFORMANCE.md` for details.

[TO FILL: the measured budget table for Pi and RGB30 is not published yet]

Where to go next

1. Read `docs/GAME-GUIDE.md` and `docs/design/games-architecture.md` to explore the template state machine.
2. Consult the *bm SDK* and *bm Code* manuals and the API sheets in `docs/api/` .
3. If anything blocks you, tell the person who gave you the card what stopped you, and where.

Open points

- **Make the game on the RGB30**: the Stages 2D template is not in bm SDK (RGB30) yet: check these steps when it is
- **Make it yours**: the Stages 2D template is not in bm SDK (RGB30) yet: check these steps when it is
- **On the Pi: change the code with bm Code**: a project made on the RGB30 opened in bm Code on the Pi: not tried yet
- **On the Pi: change the code with bm Code**: Ctrl+Q, the help on the function under the cursor in bm Code: not there yet
- **Keep it at 60 fps**: the measured budget table for Pi and RGB30 is not published yet